



US012412972B2

(12) **United States Patent**
Gustafsson et al.

(10) **Patent No.:** **US 12,412,972 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **WIDEBAND BALUN ARRANGEMENT**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 258 days.

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(21) Appl. No.: **18/265,669**

(Continued)

(22) PCT Filed: **Dec. 7, 2020**

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(86) PCT No.: **PCT/EP2020/084906**

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§ 371 (c)(1),

(2) Date: **Jun. 6, 2023**

(87) PCT Pub. No.: **WO2022/122114**

PCT Pub. Date: **Jun. 16, 2022**

(65) **Prior Publication Data**

US 2024/0079754 A1 Mar. 7, 2024

(51) **Int. Cl.**
H01P 5/10 (2006.01)

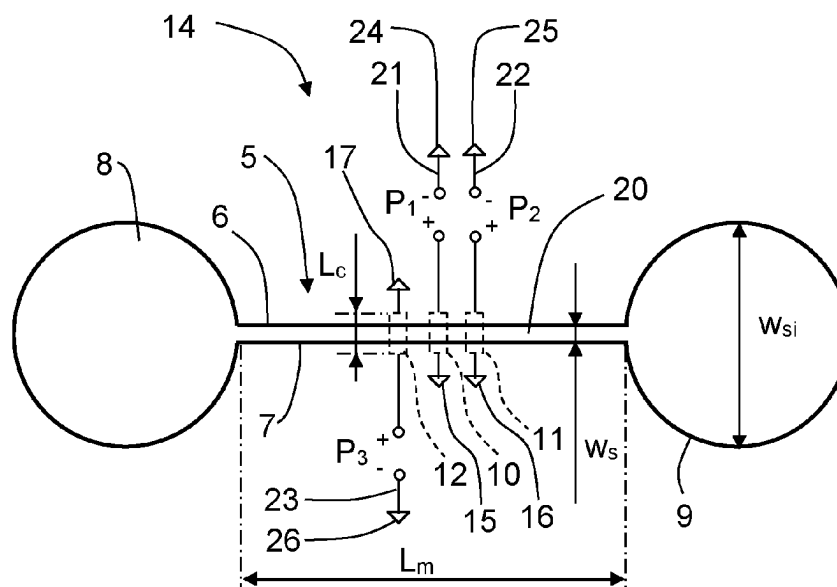
(52) **U.S. Cl.**
CPC **H01P 5/1007** (2013.01)

(58) **Field of Classification Search**
CPC H01P 5/1007; H01P 5/10
USPC 333/25, 26, 246
See application file for complete search history.

(57) **ABSTRACT**

The present disclosure relates to a balun arrangement (14) comprising a slot (5) at least partially having a crossing slot width (ws, hs) running between a first longitudinal side (6) and a second longitudinal side (7) in at least a first metalization layer (2). The balun arrangement (14) further comprises an unbalanced first port (P₁) that is defined between a first connection (10) to the second side (7) of the slot (5) and a fourth connection (21) to the first side (6) of the slot (5) and a balanced second port (P₂) that is that is defined between a second connection (11) to the second side (7) of the slot (5) and a fifth connection (22) to the first side (6) of the slot (5). The balun arrangement (14) also comprises a balanced third port (P₃) that is defined between a third connection (12) to the first side (6) of the slot (5) and a sixth connection (23) to the second side (7) of the slot (5).

17 Claims, 8 Drawing Sheets



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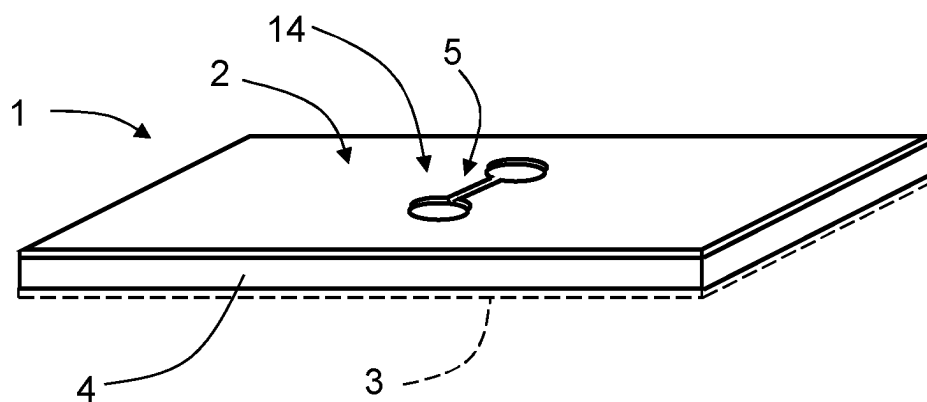


FIG. 1

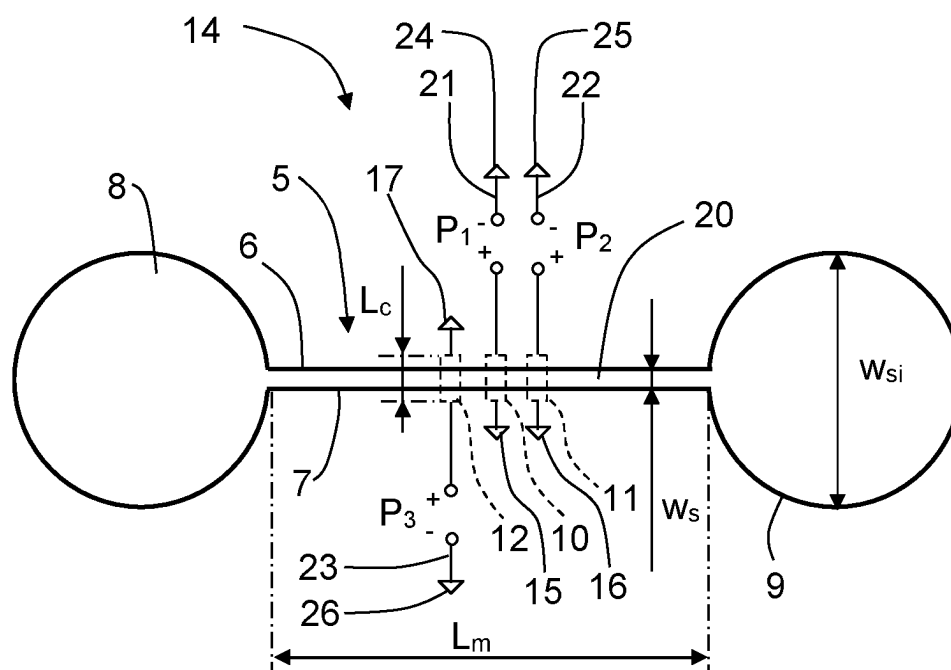


FIG. 2

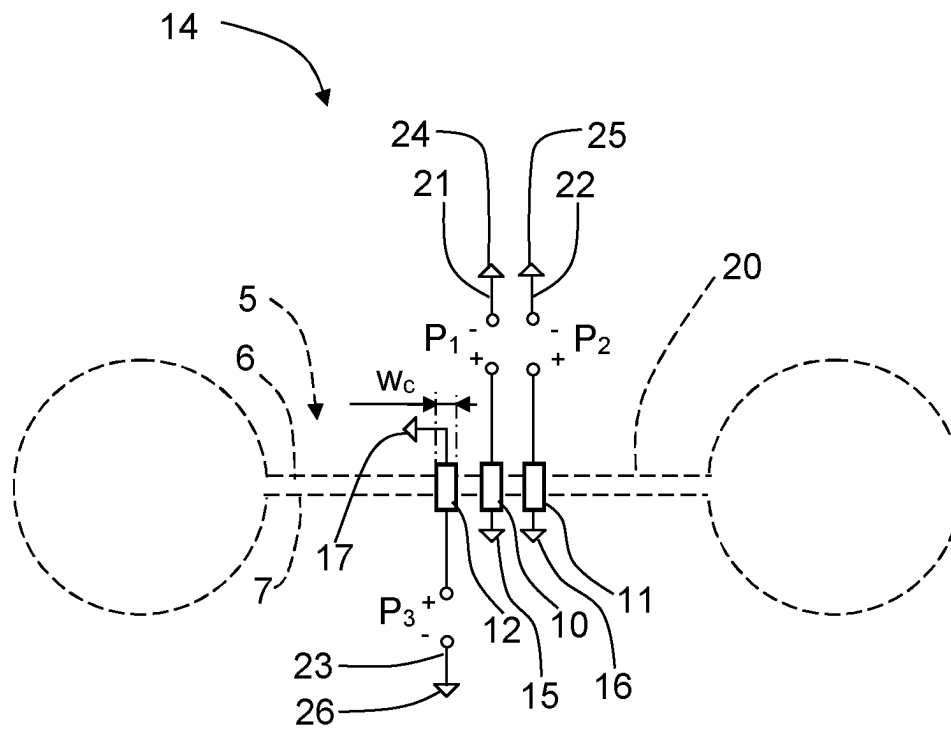


FIG. 3

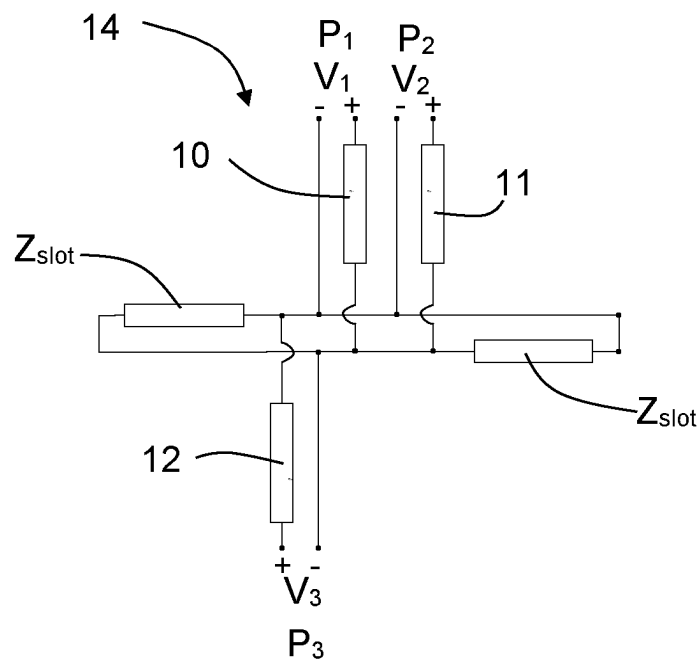


FIG. 4

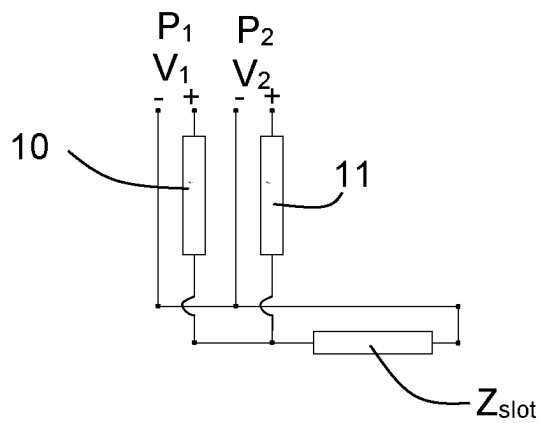


FIG. 5A

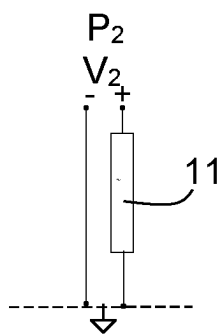


FIG. 5B

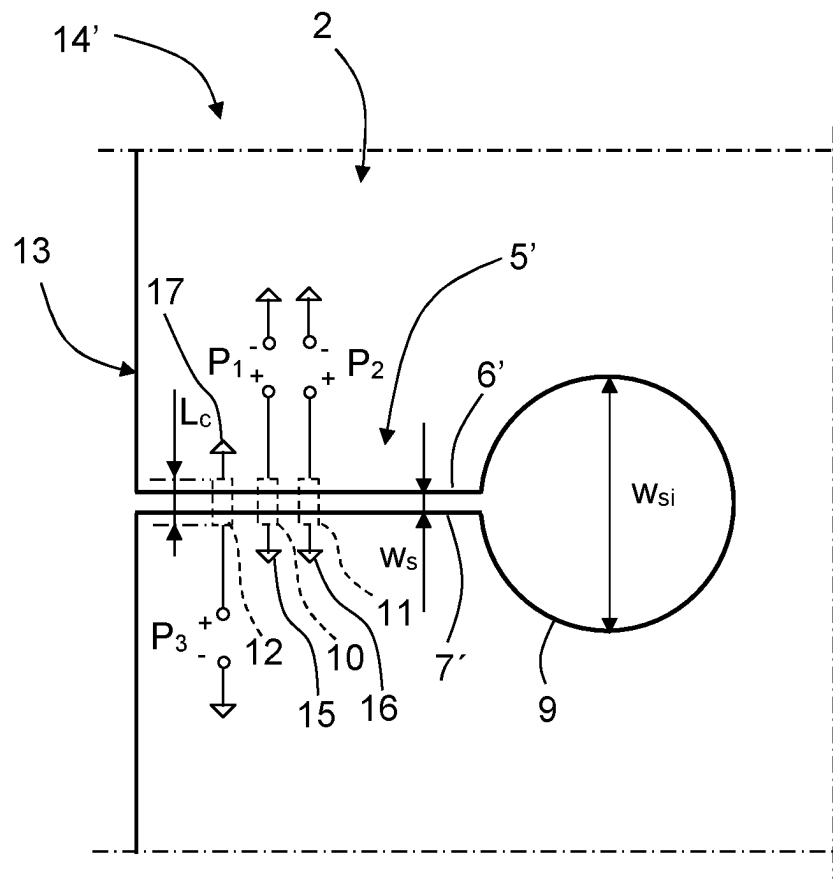


FIG. 6

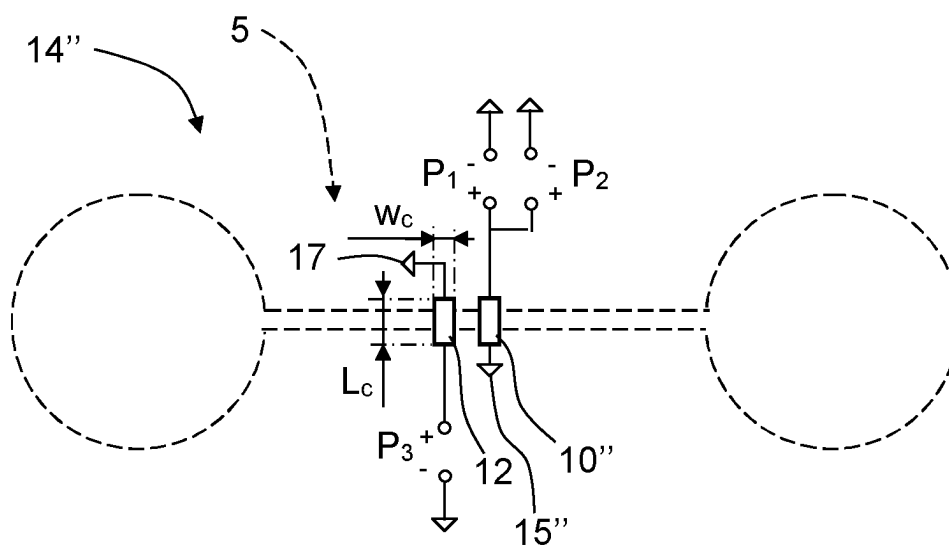


FIG. 7

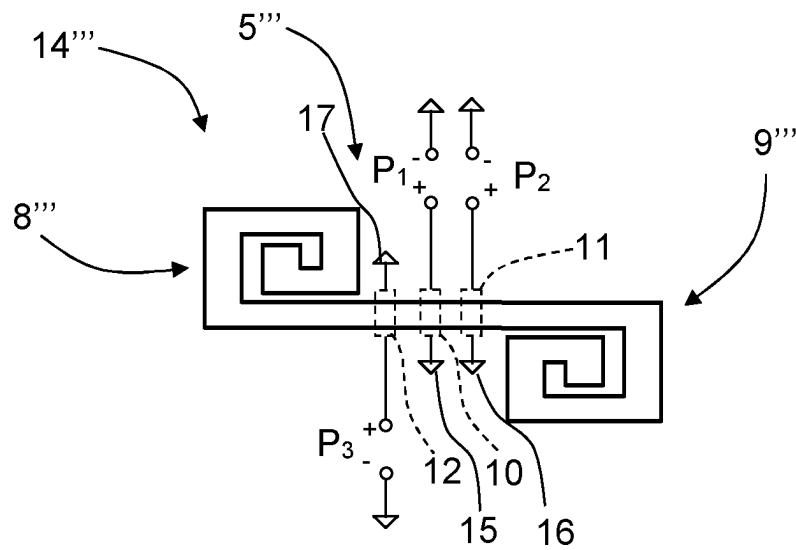


FIG. 8

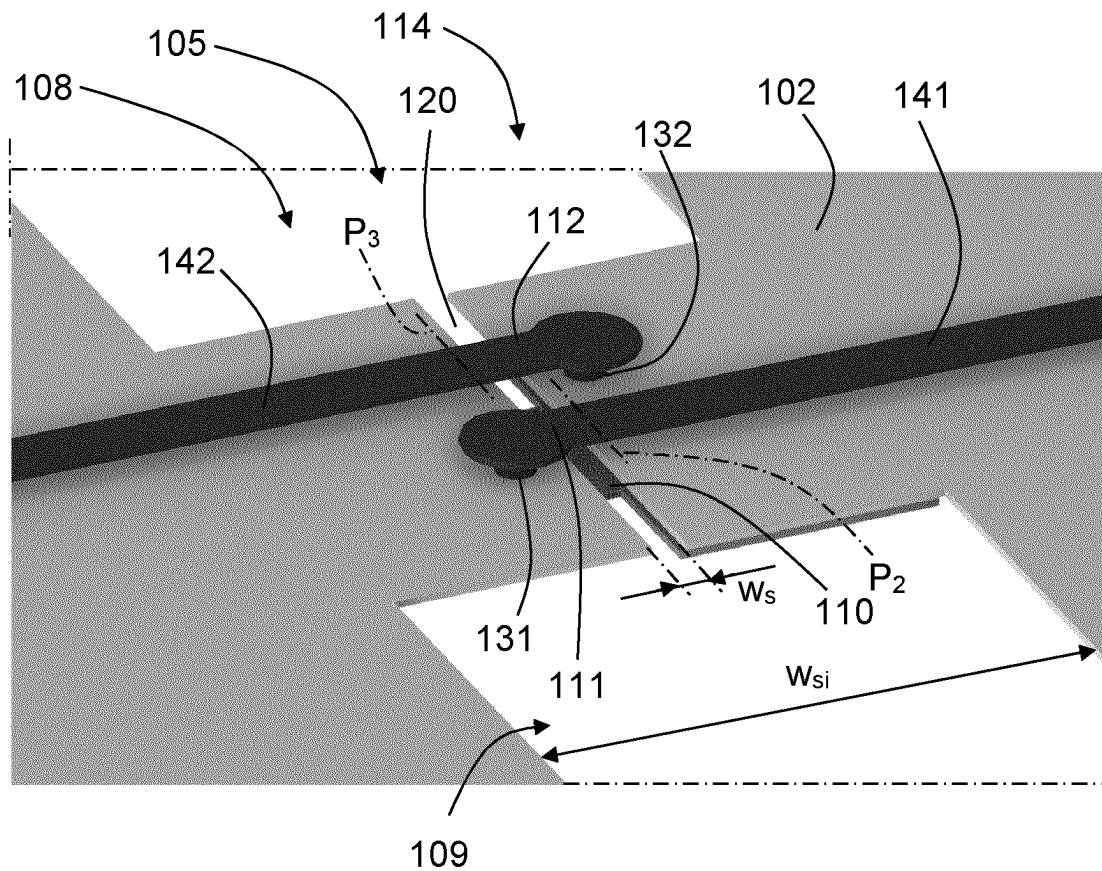


FIG. 9

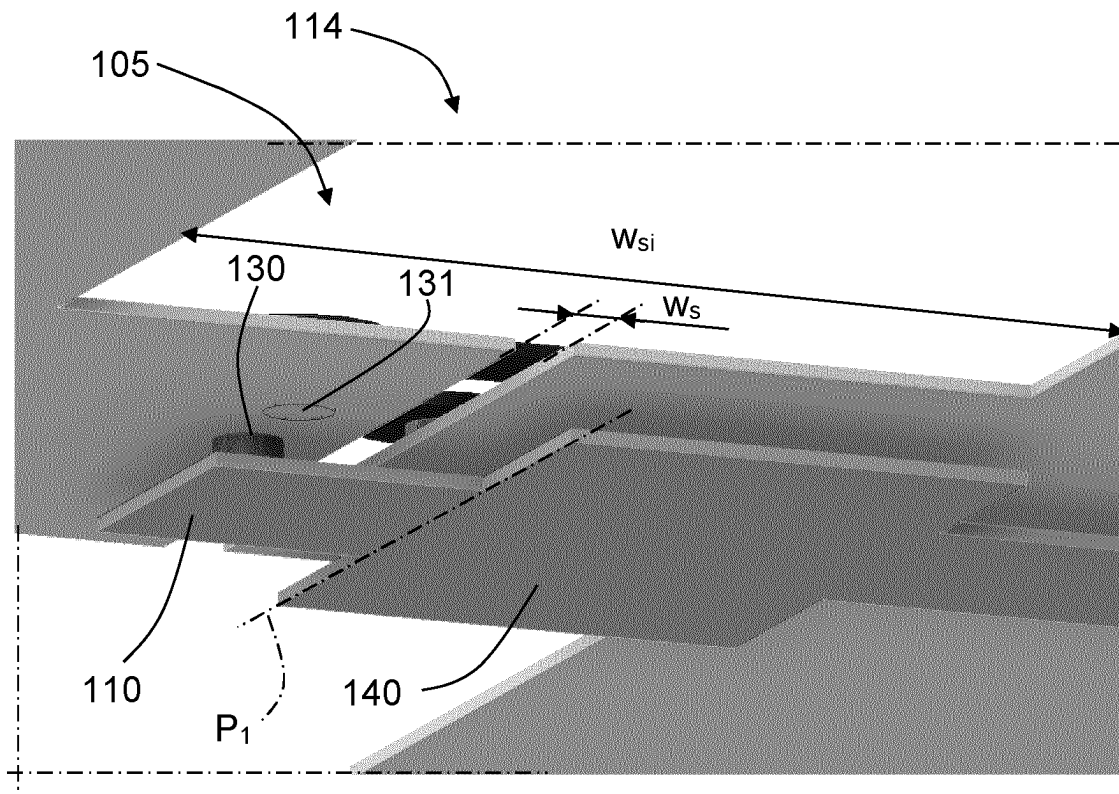


FIG. 10

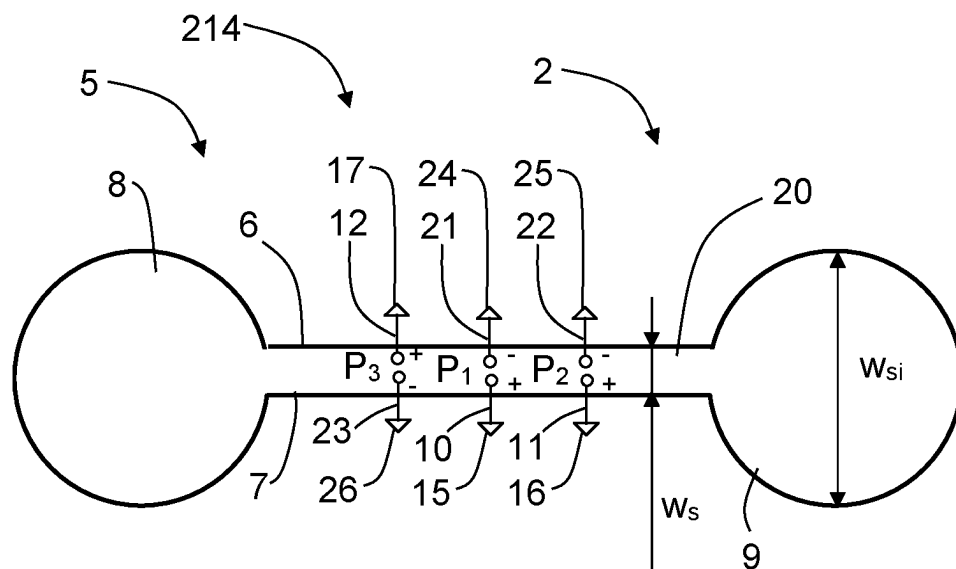


FIG. 11

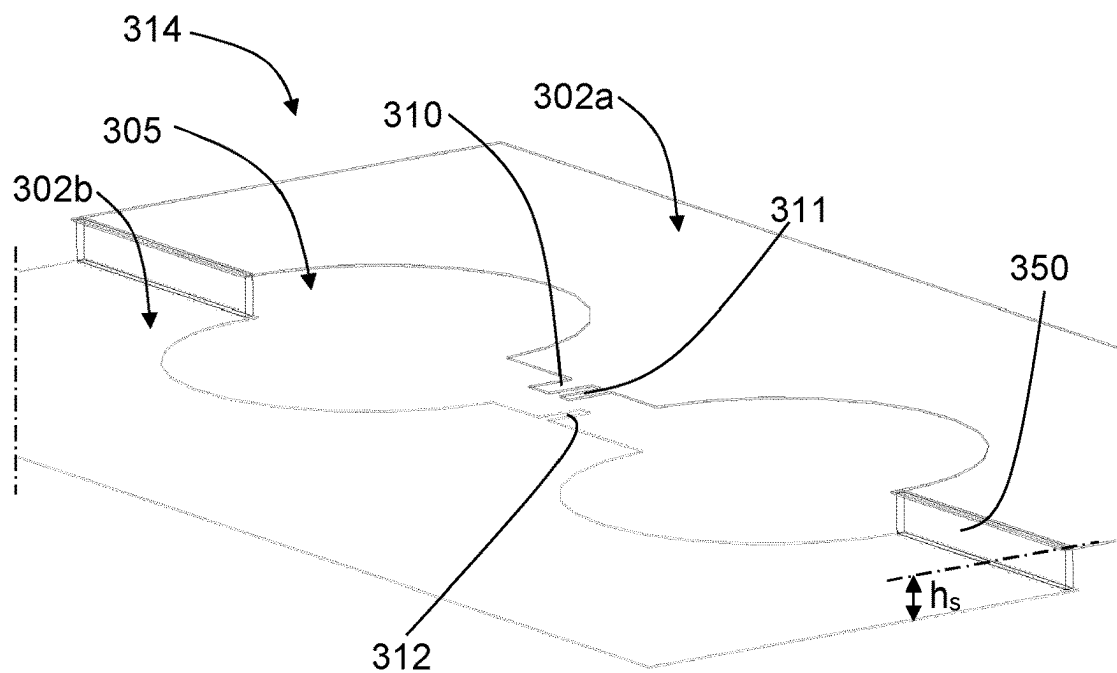


FIG. 12

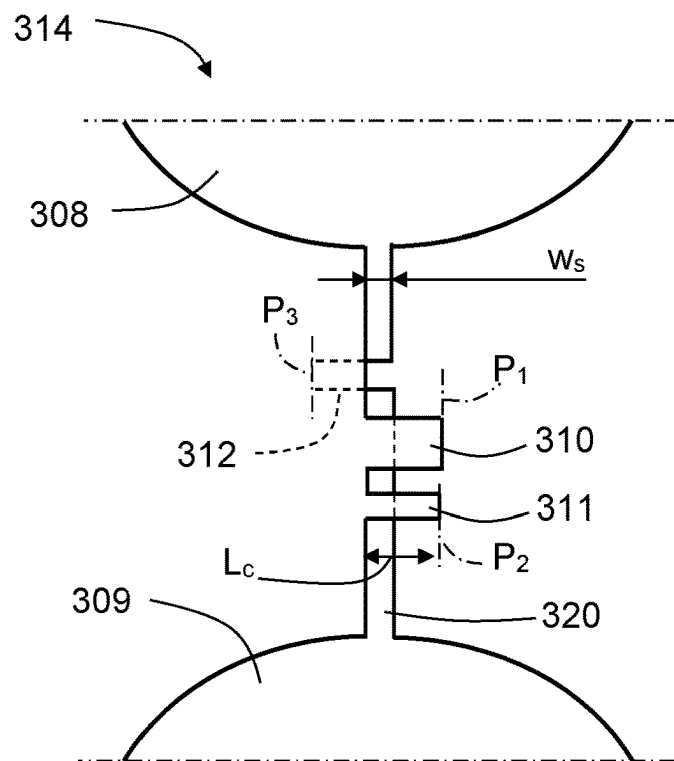
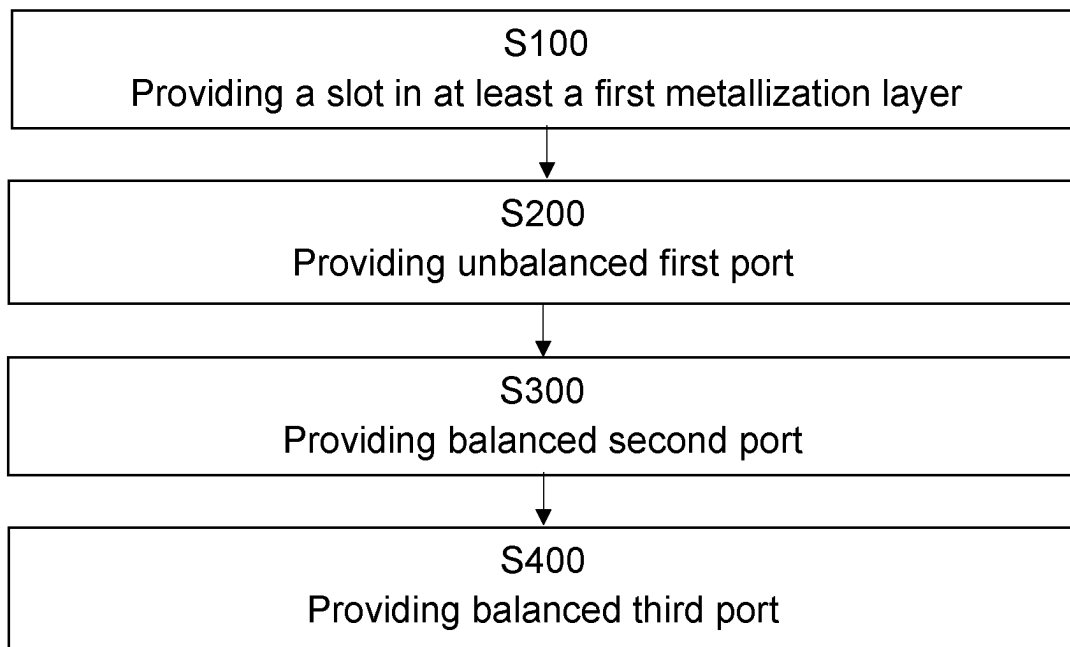


FIG. 13

FIG. 14

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WIDEBAND BALUN ARRANGEMENT**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a 35 U.S.C. § 371 National Stage of International Patent Application No. PCT/EP2020/084906, filed Dec. 7, 2020, the disclosure of which is incorporated herein in its entirety.

TECHNICAL FIELD

The present disclosure relates to balun arrangements in RF (Radio Frequency) and micro/millimeter wave circuits.

BACKGROUND

The balun is an important circuitry used in a multitude of RF and microwave circuits such as push-pull amplifiers, balanced mixers, balanced multipliers, antenna feeds etc. Desired balun features are stable phase and amplitude balance within a broad, or wide, bandwidth, and in many cases low common mode impedance is also desired. In addition, at GHz-frequencies and higher, where circuits are made in PCB-, substrate-, or RFIC/MMIC-technology, it is most often required that the balun can be realized in a planar design.

Another well-known balun topology is the so-called Guanella balun, also known as the Guanella transmission line transformer. The performance of a planar Guanella balun is based on coupled lines and its performance is dependent on the ratio between the even and odd mode impedance for its coupled lines. Ideally, for any coupled line realization with finite even mode impedance, the Guanella balun will only have 0 dB and 180° amplitude- and phase balance at the frequency where the lines are 900 long. For any other frequency there will be a non-ideal balance. For this reason, the implementation of the Guanella balun may benefit from a defected ground structure (DGS), referring to the removal of the ground plane in the vicinity of the coupled lines, effectively increasing the even to odd mode impedance ratio.

One such example is presented in “A defected-ground coupled line section with two shorts for wideband balun application” by B. Li, X. Wu, J. Yang and W. Wu, 2009 Asia Pacific Microwave Conference, Singapore, 2009, pp. 2030-2032, doi:10.1109/APMC.2009.5385296 where a thin ground plane slot passes under the center of a 90° coupled line segment, effectively increasing the even mode impedance.

A common topology used to overcome the limitations imposed by the finite even mode impedance of coupled lines is the Marchand balun, also having a previously well-known ideal topology. The ideal Marchand balun has a phase and amplitude balance that is independent of the even and odd mode impedance of the coupled line segments, thereby enabling very large bandwidths. However, the wideband phase and amplitude balance requires that the even and odd mode impedance of the coupled lines maintain stable over frequency which seldom is the case in any real implementation.

In addition, the Marchand balun relies on coupled lines being 90 long at a center frequency, having the consequence that the input and output impedance of the balun becomes frequency dependent.

A general problem with wideband baluns implemented in planar technologies at GHz frequencies is that when driven

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in common mode, the balanced ports will see an open circuit impedance, or a strongly frequency dependent reactive impedance. Although common mode operation impedance is of less importance in applications where harmonic contents is negligible, there are many applications when it's of highest importance. This is for instance the case for push-pull amplifiers where common mode operation needs to be short-circuited, or at least terminated with low impedance, at the output of the transistors in order to ensure high efficiency and linearity. Another example is balanced mixers where low common mode operation impedance can increase conversion gain.

It is therefore desired to provide a wideband balun that functions in a differential mode operation with low common mode operation impedance at the balanced ports.

SUMMARY

It is an object of the present disclosure to provide a balun arrangement that enables wideband capabilities in both common mode and differential mode operation.

This object is obtained by means of a balun arrangement comprising a slot at least partially having a crossing slot width running between a first longitudinal side and a second longitudinal side in at least a first metallization layer. The balun arrangement further comprises an unbalanced first port that is defined between a first connection to the second side of the slot and a fourth connection to the first side of the slot and a balanced second port that is that is defined between a second connection to the second side of the slot and a fifth connection to the first side of the slot. The balun arrangement also comprises a balanced third port that is defined between a third connection to the first side of the slot and a sixth connection to the second side of the slot.

This provides a compact and uncomplicated wideband balun arrangement that functions in a differential mode operation with low common mode operation impedance at the balanced ports. Since the slot is short-circuited at its ends, an increased impedance, or reduced coupling, will increase the differential mode bandwidth of the balun arrangement.

According to some aspects, the first connection runs from a first type terminal of the first port to a first ground connection at the second side of the slot, and the fourth connection runs from a second type terminal of the first port to a fourth ground connection at the first side of the slot. The second connection runs from a first type terminal of the second port to a second ground connection at the second side of the slot, and the fifth connection runs from a second type terminal of the second port to a fifth ground connection at the first side of the slot. The third connection runs from a first type terminal of the third port to a third ground connection at the first side of the slot, and the sixth connection runs from a second type terminal of the third port to a sixth ground connection at the second side of the slot.

According to some aspects, the first connection, the second connection and the third connection cross the slot, wherein the length of each connection between the corresponding port ports and ground connection is equal to, or exceed the crossing slot width and has an electrical length that falls below 20°, and preferably below 10°, at a center frequency at a desired frequency band.

By minimizing the length of each connection, the common mode impedance can be made very low, ideally short circuit when electrical length equals 0°. Simultaneously, the amplitude and phase balance in differential mode becomes wideband by a proper design of the ground plane slot.

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According to some aspects, the connections are in the form of microstrip conductors which are formed in at least one metallization layer.

This enables the balun arrangement to be realized in PCB technology.

According to some aspects, the microstrip conductors are formed in at least one metallization layer that is separate from the first metallization layer. According to some further aspects, the microstrip conductors are formed in at least two metallization layers that are separate from the first metallization layer.

This provides versatility regarding how to realize the balun arrangement.

According to some aspects, the slot comprises a maximum increased width that exceeds the crossing slot width.

This way, the coupling between opposite sides of the slot is reduced.

According to some aspects, the slot is terminated in an open circuit at an edge of a printed circuit board (PCB) that comprises the metallization layers.

This way, a more compact slot and balun arrangement is provided.

According to some aspects, the first connection and the second connection are merged into a combined connection such that the first connection and the second connection are one and the same, being connected to both the first port and the second port.

This way, a more compact slot and balun arrangement is provided.

According to some aspects, the slot is formed in a first metallization layer and a second metallization layer, such that the slot has a horizontal slot width and a vertical slot width defined by a height between the metallization layers. The metallization layers are electrically connected by vertical connections, where a respective first port and second port are defined across the slot width from the first metallization layer to the second metallization layer. A third port is defined across the slot width from the second metallization layer to the first metallization layer.

This provides versatility regarding how to realize the balun arrangement, taking advantage of a multilayer structure where the slot can be realized in different layers.

According to some aspects, the horizontal slot width w_s is zero or negative.

This way, a slot can be formed even by means of horizontally meeting or overlapping ground planes in different layer, allowing the ports to be formed at the edge of, or within, the ground planes.

According to some aspects, the horizontal slot width has a positive value, and there is a first conductor and a second conductor in the first metallization layer. The first conductor extends over the slot and ends in the first port, and the second conductor extends over the slot and ends in the second port. There is a third conductor in the second metallization layer, where the third conductor extends over the slot and ends in the third port.

This object is also obtained by means of methods that are associated with the above advantages.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure will now be described more in detail with reference to the appended drawings, where:

FIG. 1 shows a schematic perspective view of a PCB with a first example of a balun arrangement;

FIG. 2 shows a schematic first main side view of the first example of a balun arrangement;

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FIG. 3 shows a schematic second main side view of the first example of a balun arrangement;

FIG. 4 shows a simplified equivalent circuit of the first example of a balun arrangement;

FIG. 5A shows a simplified equivalent circuit of the first example of a balun arrangement in a differential mode;

FIG. 5B shows a simplified equivalent circuit of the first example of a balun arrangement in a common mode;

FIG. 6 shows a schematic first main side view of a second example of a balun arrangement;

FIG. 7 shows a schematic second main side view of a third example of a balun arrangement;

FIG. 8 shows a schematic first main side view of a fourth example of a balun arrangement;

FIG. 9 shows a schematic first perspective view of a fifth example of a balun arrangement;

FIG. 10 shows a schematic second perspective view of the fifth example of a balun arrangement;

FIG. 11 shows a schematic first main side view of a general realization of the present disclosure;

FIG. 12 shows a schematic perspective view of a sixth example of a balun arrangement;

FIG. 13 shows a schematic enlarged top view of the sixth example of a balun arrangement; and

FIG. 14 shows a flowchart for methods according to the present disclosure.

DETAILED DESCRIPTION

Aspects of the present disclosure will now be described more fully hereinafter with reference to the accompanying drawings. The different devices, systems, computer programs and methods disclosed herein can, however, be realized in many different forms and should not be construed as being limited to the aspects set forth herein. Like numbers in the drawings refer to like elements throughout.

The terminology used herein is for describing aspects of the disclosure only and is not intended to limit the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Ground is referring to RF ground, potentially via a capacitor.

With reference to FIG. 1, FIG. 2 and FIG. 3, there is a PCB 1 (printed circuit board) that comprises a first metallization layer 2, a second metallization layer 3 and an intermediate dielectric layer 4. The first metallization layer 2 constitutes a first ground plane and comprises a slot 5 having a mid section 20 with a slot width w_s , a first longitudinal side 6 and a second longitudinal side 7, where the slot 5 further has a first end part 8 with increased maximum slot width w_{si} and a second end part 9 with the increased maximum slot width w_{si} . By means of this increased maximum slot width w_{si} , the coupling between opposite sides of the slot 5 is reduced.

The slot 5 is of the type defected ground structure (DGS), an unbalanced first port P_1 is connected to ground via a first galvanic connection that runs from the first port P_1 to a first ground connection 15. According to some aspects, the first galvanic connection is in the form of a first microstrip conductor 10 having a first characteristic impedance $Z_{C,10}$ and being formed in the second metallization layer 3, on the opposite PCB side of the slot 5. The slot 5 is used to separate a balanced second port P_2 and balanced third port P_3 . The second port P_2 is connected to ground via a first galvanic connection that runs from the second port P_2 to a second ground connection 16. According to some aspects, the second galvanic connection is in the form of a second

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microstrip conductor **11** having a second characteristic impedance $Z_{C,20}$ and being formed in the second metallization layer **3**. The third port P_3 is connected to ground via a third galvanic connection that runs from the first port P_3 to a third ground connection **17**. According to some aspects, the second galvanic connection is in the form of a third microstrip conductor **12** having a third characteristic impedance $Z_{C,20}$ and also being formed in the second metallization layer **3**. The third microstrip conductor **12** is connected to ground at an opposite side of the slot **5** compared to the first microstrip conductor **10** and the second microstrip conductor **11**. The second metallization layer **3** is indicated with dashed lines in FIG. 1 since the initial metallization has been partly removed to form the microstrip conductors **10**, **11**, **12** and other traces and connections (not shown). The slot, the ports P_1 , P_2 , P_3 and the microstrip conductors **10**, **11**, **12** form a first example of a balun arrangement **14**.

According to the present disclosure, the length L_c of the microstrip conductors **10**, **11**, **12** should be kept at a minimum, at least the length L_c should equal the slot width w_s , and the length L_c should fall below an electrical length of 20° , preferably fall below an electrical length of 10° , and even more preferably fall below an electrical length of 5° at a center frequency of a desired frequency band. In case the microstrip conductors **10**, **11**, **12** are grounded by means of via connections, as will be described later, the via length is included in the length L_c of the microstrip conductors **10**, **11**, **12**.

For reasons of clarity, the length L_c and width w_c are indicated for the third microstrip conductor **12** only in the Figures, and the individual microstrip conductors **10**, **11**, **12** all have a corresponding length L_c and width w_c where the individual microstrip conductors **10**, **11**, **12** can have different lengths L_c and different widths w_c , but the constraints above are always valid. The length L_c of each galvanic connection **10**, **11**, **12** runs between the corresponding port P_1 , P_2 , P_3 and ground connection **15**, **16**, **17**.

Although the galvanic connections are illustrated as microstrip conductors **10**, **11**, **12** in FIG. 2 and FIG. 3, other connections such as bond wires can also be used. The grounding of the microstrip conductors **10**, **11**, **12** can be made by via holes or other electrical connections, according to some aspects a capacitive connection. In the case of, for example, a capacitance being connected in series with the microstrip conductors **10**, **11**, **12**, the microstrip conductors **10**, **11**, **12** do not constitute galvanic connections, but at least they constitute electric connections.

This shape of the slot **5** makes it possible to fairly accurately regard it as two parallel short-circuited transmission lines. The electrical length of these lines are preferably in the range 20° - 120° at the center frequency. Since the slot **5** is short-circuited at its ends, an increased impedance, or reduced coupling, will increase the differential mode bandwidth of the balun arrangement.

The ports P_1 , P_2 , P_3 can be connected to other components such as for example transistors, and can be used in a push-pull amplifier arrangement. The ports P_1 , P_2 , P_3 can also be connected to microstrip or stripline transmission lines, bonding wires, or similar.

The slot **5** will act as a short-circuited slot line that according to some aspects should be made as wide as possible outside the mid section **20** in order to increase the bandwidth of the balun and with a proper length for the targeted bandwidth. The increased maximum slot width w_{sl} can be different at the different end parts **8**, **9**, and the end parts can be round, as shown here, or have any other shapes such as for example square or stepped structures where the

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width increases and/or decreases in steps to a maximum value, or a tapered part where the width increases continuously to a maximum value, where the maximum value can be anywhere in the end part. The slot can have any type of shape where different end parts can have different shapes. The mid section **20** has a length L_m that should be kept as short as possible, and according to some aspects, the mid section **20** only runs where the microstrip conductors **10**, **11**, **12** cross the slot.

By minimizing the length L_c of the microstrip conductors **10**, **11**, **12**, a common mode impedance can be made very low, ideally short-circuit when the length $L_c \rightarrow 0^\circ$. Simultaneously, the amplitude and phase balance in a differential mode becomes wideband by a proper design of the ground plane slot **5**.

Here, a two-layer structure is shown where the microstrip conductors **10**, **11**, **12** are formed in a common metallization layer **3**, but the balun arrangement can also be implemented in a multilayer structure. According to some aspects, the slot can be arranged in one or more metallization layers, and the microstrip/stripline conductors can be arranged in one, two or three other metallization layer. The microstrip/stripline conductors can thus be made in the same metal layer. In the case of the slot being formed in two or more metallization layers, the conductors can run between these metallization layers.

For example, the balanced second port P_2 and third port P_3 can be connected to ground via a corresponding electrical connection, such as a microstrip/stripline conductor, in one metallization layer, and the unbalanced first port P_1 can be connected to ground via an electrical connection, such as a microstrip/stripline conductor, in another metallization layer. The metallization layers could be on different sides of the ground planes and/or be an extension of the ground plane itself.

Generally, as schematically shown in FIG. 11, the present disclosure relates to balun arrangement **214** comprising a first metallization layer **2** that in turn comprises a slot at least partially having a crossing slot width w_s , w_h running between a first side **6** and a second side **7**, where the balun arrangement **214** further comprises an unbalanced first port P_1 that is defined between a first connection **10** to the second side **7** of the slot **5** and a fourth connection **21** to the first side **6** of the slot **5**. The balun arrangement **214** also comprises a balanced second port P_2 that is that is defined between a second connection **11** to the second side **7** of the slot **5** and a fifth connection **22** to the first side **6** of the slot **5**, and a balanced third port P_3 that is defined between a third connection **12** to the first side **6** of the slot **5** and a sixth connection **23** to the second side **7** of the slot **5**.

According to some aspects, the first connection **10** runs from a positive terminal + of the first port P_1 to a first ground connection **15** at the second side **7** of the slot **5**, and the fourth connection **21** runs from a negative terminal - of the first port P_1 to a fourth ground connection **24** at the first side **6** of the slot **5**. The second connection **11** runs from a positive terminal + of the second port P_2 to a second ground connection **16** at the second side **7** of the slot **5**, and the fifth connection **22** runs from a negative terminal - of the first port P_1 to a fifth ground connection **25** at the first side **6** of the slot **5**. The third connection **12** runs from a positive terminal + of the third port P_3 to a third ground connection **17** at the first side **6** of the slot **5**, and the sixth connection **23** runs from a negative terminal - of the first port P_1 to a sixth ground connection **26** at the second side **7** of the slot **5**.

A simplified equivalent circuit of the topology for the balun arrangement **14** in FIG. 2 and FIG. 3 is shown in FIG. 4, where Z_{slot} represents the slot **5**. In the case of a having a slot with varying width and shape, as in FIG. 2 and FIG. 3, the equivalent model of the slot can be improved by cascading multiple transmission line models having varying impedances and lengths. Since the lengths L_c of the microstrip conductors **10**, **11**, **12** are relatively short, their characteristic impedance only has minor effect on the overall functionality. The short length L_c also ensures that although the microstrip conductors **10**, **11**, **12** are placed close to each other, the effect from their mutual coupling becomes low. According to some aspects, the term short refers to an electrical length of magnitudes as discussed above.

At the first port P_1 there is a first voltage V_1 , at the second port P_2 there is a second voltage V_2 , and at the third port P_3 there is a third voltage V_3 . In a differential mode, the second voltage V_2 equals the negative third voltage V_3 such that $V_2 = (-V_3)$. In a common mode, the second voltage V_2 equals the third voltage V_3 such that $V_2 = V_3$. This is illustrated in FIG. 5A and FIG. 5B that shows the equivalent circuit, as seen from the second port P_2 , when the second port P_2 and the third port P_3 are excited in differential mode and common mode, respectively.

As seen in FIG. 5A, at differential mode, the circuit simplifies to a shunt stub Z_{slot} , assuming that the microstrip conductors **10**, **11**, **12** are relative short. As seen in FIG. 5B, at common mode the circuit is a short transmission line connected to a virtual ground. Hence, the desired wideband low impedance at common mode is achieved when the second microstrip conductor **11** and the third microstrip conductor **12** are relatively short. The differential mode will experience a frequency dependence due to the short-circuited stub Z_{slot} implying that a larger impedance of Z_{slot} improves the bandwidth.

FIG. 6 shows a second example of a balun arrangement **14'** where the left side of the slot **5'** is terminated in an open circuit at an edge **13** of the circuit board **1**, resulting in a potentially larger differential mode bandwidth compared to the configuration in FIG. 2 and FIG. 3.

FIG. 7 shows a third example of a balun arrangement **14''**. In FIG. 2 and FIG. 3, the first microstrip conductor **10** and the second microstrip conductor **11** are connected in parallel. In this example, they are merged into a combined microstrip conductor **10''** across the slot **5**.

It should be noted that the shape of the slot may take many different forms. For instance, FIG. 8 shows a fourth example of a balun arrangement **14'''** that comprises a folded slot-line **5'''** that is suitable when compact size is important.

Another design example of the balun arrangement according to the present disclosure is shown in FIG. 9 and FIG. 10 that show perspective views of a fifth example of a balun arrangement **114** where a three metallization layer substrate is used.

The first conductor **110** runs on one side of the first metallization layer **102**, and the second conductor **111** and the third conductor **112** run on an opposite side of the first metallization layer **102**. The ports P_1 , P_2 , P_3 are defined between an end of the respective conductor **110**, **111**, **112**, and the first metallization layer **102**. Each port P_1 , P_2 , P_3 is in this example connected to a continuing corresponding first connecting conductor **140**, second connecting conductor **141** and third connecting conductor **142** that runs in the same metallization layer as the corresponding conductor **110**, **111**, **112**. The first conductor **110** is connected to ground in the first metallization layer **102** by means of a first via connection **130** at the other side of the slot **105** with

reference to the first port P_1 . The second conductor **111** is connected to ground in the first metallization layer **102** by means of a second via connection **131** at the other side of the slot **105** with reference to the second port P_2 . The third conductor **112** is connected to ground in the first metallization layer **102** by means of a third via connection **132** at the other side of the slot **105** with reference to the third port P_3 . The first via connection **130** and the second via connection **131** are on one side of the slot **105**, and the third via connection **131** is on the other side of the slot **105**. As shown in FIG. 10, the first connecting conductor **140** has a varying width, for example for providing an appropriate matching to the first port P_1 .

In this example, the end parts **108**, **109** having the increased maximum slot width w_{st} are square, but can of course have any suitable shape. The slot **105** has a mid section **120** with the slot width w_s .

Another design example of the balun arrangement according to the present disclosure is shown in FIG. 12 and FIG. 13 that show a perspective view and an enlarged top view of a sixth example of a balun arrangement **314** where a slot **305** that is formed in two different metallization layers **302a**, **302b** is used.

In a first metallization layer **302a** there is a first conductor **310** and a second conductor **311** which extend over the slot **305** that has a horizontal slot width w_s . The conductors **310**, **311** end in a respective first port P_1 and second port P_2 and are defined between the conductors' ends and a second metallization layer **302b**. In the second metallization layer **302b** there is a third conductor that extends over the slot **305** and ends in a third port P_3 that is defined between the conductor's end and the first metallization layer **302a**.

A height h_s between the metallization layers **302a**, **302b** defines a vertical slot width h_s where the slot **305** is partially formed in the first metallization layer **302a** and the second metallization layer **302b**, being longitudinally divided over the slot gap that defines the slot width w_s for a mid section **320** of the slot **305**. The metallization layers **302a**, **302b** are electrically connected by vertical connections **350** such as for example vias or metal platings.

This means that the slot **305** that is formed in two different metallization layers **302a**, **302b**, and the ports P_1 , P_2 , P_3 are also formed in these layers **302a**, **302b**. The length L_c of the conductors **310**, **311**, **321** should be sufficient to reach over the slot **305** across the horizontal slot width w_s . According to some aspects, the horizontal slot width w_s can be reduced to zero, such that the slot **305** only has the vertical slot width h_s . In the case, the conductors **310**, **311**, **321** are not needed and the length L_c is zero, and the ports P_1 , P_2 , P_3 can be formed directly at the longitudinal sides **106**, **107**. As mentioned previously, by minimizing the length L_c of the conductors **310**, **311**, **312**, a common mode impedance can be made very low, ideally short-circuit when the length $L_c \rightarrow 0^\circ$.

Furthermore, it is conceivable that the different metallization layers **302a**, **302b** and thus the longitudinal sides **106**, **107** overlap, and in that case the horizontal slot width w_s could be regarded as negative, and also the length L_c could be regarded as negative. Then the ports P_1 , P_2 , P_3 can be formed in the corresponding metallization layer **302a**, **302b**, a certain distance from the corresponding longitudinal side **106**, **107**.

In other words, the ports P_1 , P_2 , P_3 can be formed at the edge of, or within, the ground planes **302a**, **302b**.

It is to be noted that the ports P_1 , P_2 , P_3 and ground connections only are schematically indicated in the Figures, having a well-known characteristics. For example, the ports

P_1 , P_2 , P_3 are adapted to enable an electrical connection to another component or a conductor of any type. The ground connections are adapted to provide a connection to a ground plane or similar, for example by means of a via connection to ground plane in another metallization layer.

A desired frequency band is a frequency band for which the balun arrangement is intended to be operational.

The balun arrangement can be realized in other techniques than PCB, for example the metallization layers can be formed as metal sheet parts that are separated by a dielectric spacing material that can be solid, liquid or gas, for example a foam material is possible. 3D-printed layered metal structures are also conceivable, where two or more layers, and even all layers, can be coherently formed.

With reference to FIG. 14, the present disclosure also relate to a method for configuring a balun arrangement 140, where the method comprises providing S100 a slot 5 at least partially having a crossing slot width w_s , h_s running between a first longitudinal side 6 and a second longitudinal side 5 in at least a first metallization layer 2. The method further comprises providing S200 an unbalanced first port P_1 that is defined between a first connection 10 to the first side 6 of the slot 5 and a fourth connection 21 to the second side 7 of the slot 5 and providing S300 a balanced second port P_2 that is defined between a second connection 11 to the first side 6 of the slot 5 and a fifth connection 22 to the second side 7 of the slot 5. The method further comprises providing S400 a balanced third port P_3 that is defined between a third connection 12 to the first side 6 of the slot 5 and a sixth connection 23 to the second side 7 of the slot 5.

According to some aspects, the first connection 10, the second connection 11 and the third connection 12 cross the slot 5, wherein the length L_c of each connection 10 between the corresponding port ports P_1 , P_2 , P_3 and ground connection 15, 16, 17 is equal to, or exceed the crossing slot width w_s and has an electrical length that falls below 20° , and preferably below 10° , at a center frequency at a desired frequency band.

According to some aspects, the connections are in the form of microstrip conductors 10, 11, 12 which are provided in at least one metallization layer 3.

According to some aspects, the microstrip conductors 10, 11, 12 are formed in at least one metallization layer 3 that is separate from the first metallization layer 2.

According to some aspects, the microstrip conductors 110, 111, 112 are provided in at least two metallization layers that are separate from the first metallization layer 102.

According to some aspects, the slot 5 has a maximum increased width w_{si} that exceeds the crossing slot width w_s , w_h .

According to some aspects, the slot 5' is terminated in an open circuit at an edge 13 of a printed circuit board, PCB, 1 that comprises the metallization layers 2, 3.

According to some aspects, the first connection and the second connection are merged into a combined connection 10" such that the first connection and the second connection are one and the same, being connected to both the first port P_1 and the second port P_2 .

According to some aspects, the slot 5''' is a folded slot-line 5''' that comprises at least one folded part 8''', 9''' where the slot 5''' extends in different directions

According to some aspects, the slot 305 is formed in a first metallization layer 302a and a second metallization layer 302b, such that the slot 305 has a horizontal slot width w_s and a vertical slot width h_s defined by a height h_s between the metallization layers 302a, 302b. The metallization layers 302a, 302b are electrically connected by vertical connec-

tions 350, where a respective first port P_1 and second port P_2 are defined across the slot width w_s , h_s from the first metallization layer 302a to the second metallization layer 302b. A third port P_3 is defined across the slot width w_s , h_s from the second metallization layer 302b to the first metallization layer 302a.

According to some aspects, the horizontal slot width w_s is zero or negative, where a negative slot width w_s in this context corresponds to the metallizations layers or ground planes forming the slot being horizontally overlapping and vertically separated, i.e. being positioned in different layers in a multi-layer structure. When the horizontal slot width w_s is zero, the metallizations layers or ground planes forming the slot are just about to horizontally overlap while being vertically separated. According to some aspects, in this context, the term horizontal means along the planes of the metallizations layers, perpendicular to being vertical.

According to some aspects, the horizontal slot width w_s has a positive value, and there is a first conductor 310 and a second conductor 311 in the first metallization layer 302a, where the first conductor 310 extends over the slot 305 and ends in the first port P_1 . The second conductor 311 extends over the slot 305 and ends in the second port P_2 . There is a third conductor 312 in the second metallization layer 302b, where the third conductor 312 extends over the slot 305 and ends in the third port P_3 .

The present disclosure is not limited to the above, but may vary freely within the scope of the appended claims. For example, according to some aspects, the slot can have many other shapes than the ones disclosed.

In the present context, the PCB does not have to be a traditional PCB, but can be any layered structure such as for example MMIC (Monolithic Microwave Integrated Circuit), RFIC (Radio-Frequency Integrated Circuit), substrate, etc.

The positive terminals + and the negative terminals - can change place in all examples, and generally there is a first type terminal that can be either positive or negative, and a second type terminal that has a reversed polarity compared to the first type terminal.

According to some aspects, the ground symbols in the drawings denote a local ground or local reference plane, where the slot separates the grounds.

The invention claimed is:

1. A balun arrangement comprising a slot at least partially having a crossing slot width (w_s , h_s) running between a first longitudinal side and a second longitudinal side in at least a first metallization layer, where the balun arrangement further comprises:

- a) an unbalanced first port (P_1) that is defined between a first electrical connection to the second longitudinal side of the slot and a fourth electrical connection to the first longitudinal side of the slot,
- b) a balanced second port (P_2) that is defined between a second electrical connection to the second longitudinal side of the slot and a fifth electrical connection to the first longitudinal side of the slot, and
- c) a balanced third port (P_3) that is defined between a third electrical connection to the first longitudinal side of the slot and a sixth electrical connection to the second longitudinal side of the slot, wherein:

the first electrical connection runs from a first type terminal (+) of the first port (P_1) to a first ground connection at the second side of the slot, and the fourth electrical connection runs from a second type terminal (-) of the first port (P_1) to a fourth ground connection at the first longitudinal side of the slot,

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the second electrical connection runs from a first type terminal (+) of the second port (P_2) to a second ground connection at the second side of the slot, and the fifth electrical connection runs from a second type terminal (-) of the second port (P_2) to a fifth ground connection at the first longitudinal side of the slot, and

the third electrical connection runs from a first type terminal (+) of the third port (P_3) to a third ground connection at the first longitudinal side of the slot, and the sixth electrical connection runs from a second type terminal (-) of the third port (P_3) to a sixth ground connection at the second side of the slot.

2. The balun arrangement according to claim 1, wherein the electrical connections are in the form of microstrip conductors which are formed in at least one further metallization layer.

3. The balun arrangement according to claim 2, wherein the microstrip conductors are formed in the at least one further metallization layer that is separate from the first metallization layer.

4. The balun arrangement according to claim 3, wherein the microstrip conductors are formed in at least two further metallization layers that are separate from the first metallization layer.

5. The balun arrangement according to claim 1, wherein the first electrical connection, the second electrical connection and the third electrical connection cross the slot, wherein a length (L_C) of each electrical connection between the corresponding port ports (P_1 , P_2 , P_3) and ground connection is equal to, or exceed the crossing slot width (w_s).

6. The balun arrangement according to claim 1, wherein the slot is formed in the first metallization layer and a second metallization layer, such that the slot has a horizontal slot width (w_s) and a vertical slot width (h_s) defined by a height between the first and second metallization layers, where the first and second metallization layers are electrically connected by vertical electrical connections, where the first port (P_1) and the second port (P_2) are defined across the slot width (w_s , h_s) from the first metallization layer to the second metallization layer, and where the third port (P_3) is defined across the slot width (w_s , h_s) from the second metallization layer to the first metallization layer.

7. The balun arrangement according to claim 1, wherein the slot comprises a maximum increased width (w_{st}) that exceeds the crossing slot width (w_s , w_h).

8. The balun arrangement according to claim 1, wherein the slot is terminated in an open circuit at an edge of a printed circuit board, PCB, that comprises the first metallization layer.

9. The balun arrangement according to claim 1, wherein the slot is a folded slot-line that comprises at least one folded part where the slot extends in different directions.

10. A method for configuring a balun arrangement, the method comprising:

providing a slot at least partially having a crossing slot width (w_s , h_s) running between a first longitudinal side and a second longitudinal side in at least a first metallization layer;

providing an unbalanced first port (P_1) that is defined between a first electrical connection to the second longitudinal side of the slot and a fourth electrical connection to the first longitudinal side of the slot;

providing a balanced second port (P_2) that is defined between a second electrical connection to the second longitudinal side of the slot and a fifth electrical connection to the first longitudinal side of the slot; and

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providing a balanced third port (P_3) that is defined between a third electrical connection to the first longitudinal side of the slot and a sixth electrical connection to the second longitudinal side of the slot, wherein:

the first electrical connection runs from a first type terminal (+) of the first port (P_1) to a first ground connection at the second longitudinal side of the slot, and the fourth electrical connection runs from a second type terminal (-) of the first port (P_1) to a fourth ground connection at the first longitudinal side of the slot,

the second electrical connection runs from a first type terminal (+) of the second port (P_2) to a second ground connection at the second longitudinal side of the slot, and the fifth electrical connection runs from a second type terminal (-) of the second port (P_2) to a fifth ground connection at the first longitudinal side of the slot, and

the third electrical connection runs from a first type terminal (+) of the third port (P_3) to a third ground connection at the first longitudinal side of the slot, and the sixth electrical connection runs from a second type terminal (-) of the third port (P_3) to a sixth ground connection at the second longitudinal side of the slot.

11. The method according to claim 10, wherein the electrical connections are in the form of microstrip conductors which are provided in at least one further metallization layer.

12. The method according to claim 11, wherein the microstrip conductors are formed in the at least one further metallization layer that is separate from the first metallization layer.

13. The method according to claim 12, wherein the microstrip conductors are provided in at least two further metallization layers that are separate from the first metallization layer.

14. The method according to claim 10, wherein the slot has a maximum increased width (w_{st}) that exceeds the crossing slot width (w_s , w_h).

15. The method according to claim 10, wherein the first electrical connection, the second electrical connection and the third electrical connection cross the slot, wherein a length (L_C) of each electrical connection between the corresponding port ports (P_1 , P_2 , P_3) and ground connection is equal to, or exceed the crossing slot width (w_s).

16. The method according to claim 10, wherein the slot is terminated in an open circuit at an edge of a printed circuit board, PCB, that comprises the first metallization layer.

17. A balun arrangement comprising a slot at least partially having a crossing slot width (w_s , h_s) running between a first longitudinal side and a second longitudinal side in at least a first metallization layer, where the balun arrangement further comprises:

an unbalanced first port (P_1) that is defined between a first electrical connection to the second longitudinal side of the slot and a fourth electrical connection to the first longitudinal side of the slot,

a balanced second port (P_2) that is defined between a second electrical connection to the second longitudinal side of the slot and a fifth electrical connection to the first longitudinal side of the slot, and

a balanced third port (P_3) that is defined between a third electrical connection to the first longitudinal side of the slot and a sixth electrical connection to the second longitudinal side of the slot,

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wherein the slot is terminated in an open circuit at an edge of a printed circuit board, PCB, that comprises the first metallization layer.

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